

JONEIL CAOILE

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PROFESSIONAL SUMMARY

M.S. Biomedical Engineering candidate (Stevens, 3.9 GPA) with a B.S. in Kinesiology and 400+ hours of clinical experience across orthopedic, pediatric, and post-acute settings. Bridges clinical fluency with engineering rigor to translate device technology into language clinicians can act on. Seeking a clinical specialist role supporting device training, in-service education, and case-side troubleshooting.

TECHNICAL SKILLS

Clinical Knowledge & Patient Care: Anatomy, biomechanics, gait and balance analysis, neuromuscular and cardiopulmonary physiology, movement disorders, EMG signal analysis, hands-on Neubie electrical stimulation device, DBS / neuromodulation concepts.

Customer-Facing & Communication: Clinician collaboration, technical communication, training material development, translating technical outputs into clinician-actionable insights, presentation to clinical and faculty audiences, customer service from hospitality background.

Technical Foundation: Python (NumPy, SciPy, Matplotlib), MATLAB, R, SQL (SQLite, PostgreSQL), Excel, Qualtrics, motion capture (OptiTrack), force plates, SolidWorks, simulated-use workflow design, protocol development, IRB/HIPAA exposure (coursework).

EDUCATION

Stevens Institute of Technology , <i>M.S. Biomedical Engineering</i> Concentration: Medical Devices & Clinical Development Provost Master's Scholarship	Expected May 2026 GPA 3.9
University of St. Augustine for Health Sciences , <i>Doctor of Physical Therapy Program</i>	Sep – Dec 2023
California State University, Long Beach , <i>B.S. Kinesiology: Exercise Science</i>	Dec 2020 Cum Laude

PROJECTS

Chest Drain Air Leak Monitor – Inline Retrofit Cartridge (Advisor: Dr. Jinho Kim)	Jan 2026 – May 2026 (Expected)
- Designed the bedside LCD output to translate real-time pressure and airflow into the four-tier severity vocabulary clinicians already use. - Authored the Class II 510(k) regulatory-pathway analysis (Medela Thopaz+ predicate) and supported bench testing protocols.	
Biofeedback Musculoskeletal Rehab Study (Study & Iteration Lead)	Jan 2025 – May 2025
- Led a motion-capture study testing real-time audio feedback (Max/MSP) on knee alignment; operated OptiTrack and analyzed EMG plus joint angle data from 3 athletes. - Presented findings to faculty in accessible, non-technical language and designed visualizations of muscle activation patterns and movement quality to support clinician decision-making.	
Clinician-Facing Reliability Tool: Fall-Risk Assessment (Advisor: Dr. Antonia Zaferiou)	Jan 2025 – May 2025
- Built a SQLite-backed validation tool that cleaned a 77-patient clinical survey dataset and computed inter-rater reliability metrics (Krippendorff's Alpha, percent agreement) for a clinician-facing fall-risk assessment workflow. - Designed a clinician-facing decision-support dashboard, iterated outputs with faculty stakeholders, and translated technical reliability scores into language non-technical clinicians could act on.	

EXPERIENCE

Graduate Researcher <i>Stevens Institute of Technology</i>	Jan 2025 – Present
Design test protocols and instrumentation for the chest drain sensor module capstone in Dr. Jinho Kim's lab, including bedside readout logic for clinician end-users (Jan 2026 – Present).	
Physical Therapy Aide <i>Various Physical Therapy Clinics (Orthopedic, Pediatric, Geriatric)</i>	2016 – 2023 400+ Hours
Assisted therapists across orthopedic, pediatric, and post-acute settings with patient sessions, device setup, and treatment workflow; hands-on with the Neubie electrical stimulation device in outpatient orthopedic rehab.	

CERTIFICATIONS

Microsoft / LinkedIn Career Essentials (Apr 2026): Data Analysis, Business Analysis (95%), Project Management; Atlassian Agile Project Management (94%); Aha! Product Management Certified.